

Friction

- ❑ CHARACTERISTICS
- ❑ PROBLEMS



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ME101 (Spring 2017)

Friction

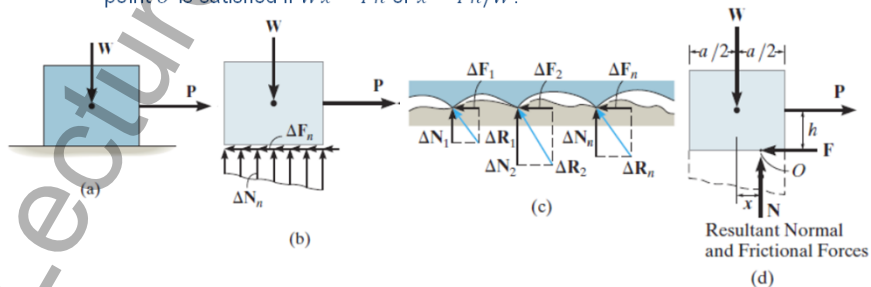
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CHARACTERISTICS OF DRY FRICTION

Definition: Friction is a force that resists the movement of two contacting surfaces that slide relative to one another.

Theory: (a) Block of uniform weight W residing on a rough horizontal surface (non-rigid)
 (b) FBD of the block (rigid)
 (c) Close Examination of how these frictional and normal forces develop
 (d) Final FBD with resultants of $\mathbf{F}$ and $\mathbf{N}$ with x to balance the tipping effect caused by $\mathbf{P}$ (centroid/geometric center of $\mathbf{N}$).
 e.g. If $\mathbf{P}$ is applied at a height h from the surface, then moment equilibrium about point O is satisfied if $Wx = Ph$ or $x = Ph/W$.



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CHARACTERISTICS OF DRY FRICTION

Impending Motion:
As P is slowly increased, F correspondingly increases until it attains a certain maximum value F_s called the *limiting static frictional force*. (*unstable equilibrium*)

$$F_s = \mu_s N$$

Motion:
If the magnitude of P acting on the block is increased so that it becomes slightly greater than F_s , the frictional force at the contacting surface will drop to a smaller value F_k called the *kinetic frictional force*.

$$F_k = \mu_k N$$

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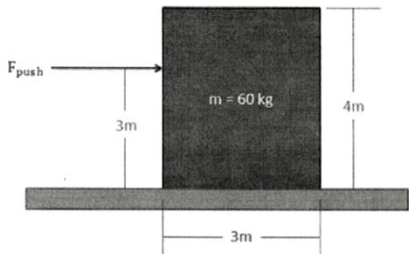
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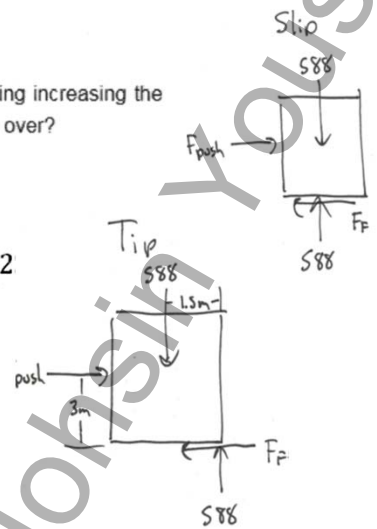
CHARACTERISTICS OF DRY FRICTION

Question 1:

The box shown below is pushed as shown. If we keep increasing the pushing force, will the box first begin to slide or will it tip over?



$m = 60 \text{ kg}$
 3 m
 4 m
 3 m
 $\mu_s = .62$



Slip
 588
 F_{push}
 F_f
 588
Tip
 588
 1.5 m
 3 m
 F_{push}
 F_f
 588

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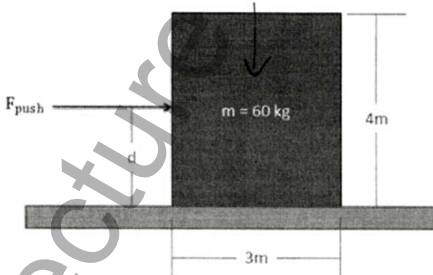
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CHARACTERISTICS OF DRY FRICTION

Question 2:

What is the maximum value of d that will allow the box to slide along the surface before tipping?



$m = 60 \text{ kg}$
 4 m
 3 m
 $\mu_s = .62$

Slip
 $F_{\text{push}} = F_f = (\mu_s)(F_N)$
 $F_{\text{push}} = 364.6 \text{ N}$

$(F_{\text{push}})(d) = (1.5)(F_N)$
 $F_{\text{push}} = \frac{(1.5)(588)}{d}$

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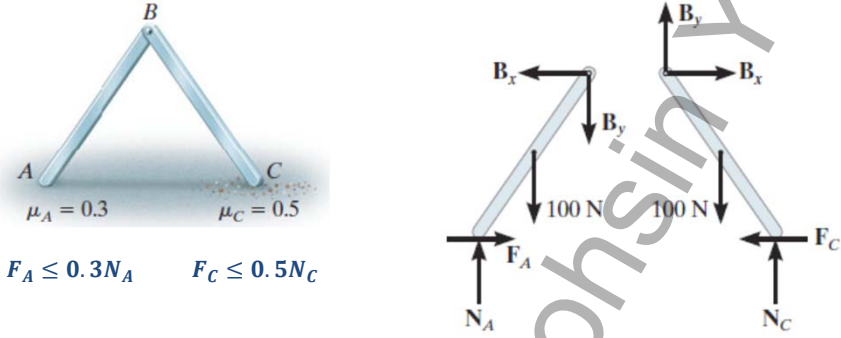
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PROBLEMS INVOLVING DRY FRICTION

Types of Friction Problems:

1. No Apparent Impending Motion

$F \leq \mu_s N$



$\mu_A = 0.3$ $\mu_C = 0.5$

$F_A \leq 0.3N_A$ $F_C \leq 0.5N_C$

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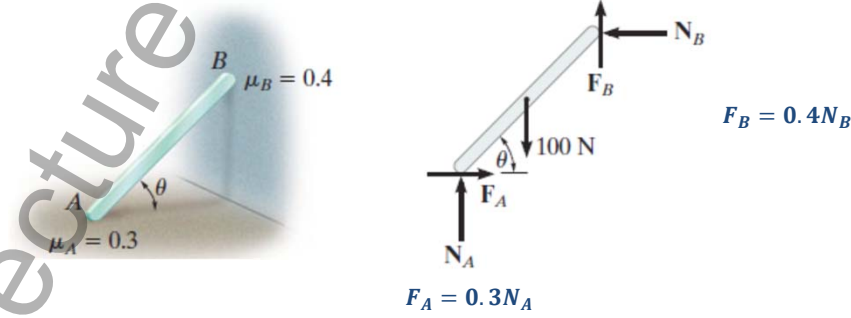
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PROBLEMS INVOLVING DRY FRICTION

Types of Friction Problems:

2. Impending Motion at all points of Contact

$F_s = \mu_s N$ and $F_k = \mu_k N$



$\mu_B = 0.4$

$\mu_A = 0.3$

$F_B = 0.4N_B$

$F_A = 0.3N_A$

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Types of Friction Problems:

3. Impending Motion at some points of Contact

$F_s = \mu_s N$ and $F_k = \mu_k N$

The diagram shows a ladder ABC with point A on the left, B at the top, and C on the right. A horizontal force P is applied at point C. The coefficient of friction at A is $\mu_A = 0.3$ and at C is $\mu_C = 0.5$. To the right, two free-body diagrams are shown. The first diagram shows the ladder with forces: N_A (normal) and F_A (friction) at A, 100 N (weight) acting downwards from the center, and N_C (normal) and F_C (friction) at C. Reaction components B_x and B_y are shown at the top joint. The second diagram shows the same forces but with the friction forces F_A and F_C acting in the opposite direction to the first diagram.

$F_A = 0.3 N_A$
 $F_C = 0.5 N_C$

$F_A \leq 0.3 N_A$
 $F_C = 0.5 N_C$

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PROBLEMS INVOLVING DRY FRICTION

The uniform crate shown has a mass of 20 kg. If a force $P = 80\text{ N}$, is applied to the crate, determine if it remains in equilibrium. The coefficient of static friction is $\mu_s = 0.3$.

The diagram shows a rectangular crate with a width of 0.8 m and a height of 0.2 m. A force P = 80 N is applied at the bottom-left corner at an angle of 30 degrees to the horizontal. A coordinate system (x, y) is shown to the right of the crate.

The free-body diagram shows the crate with forces: P = 80 N at the bottom-left corner, weight W = 20 x 9.81 N acting downwards from the center, normal force N acting upwards at the bottom-right corner, and friction force F_s acting to the right at the bottom-right corner. Dimensions are given: 0.4 m from the left edge to the point of application of N and F_s, and x m from that point to the right edge. The weight W is shown as 20 x 9.81 N.

$\vec{P} = \begin{bmatrix} 80 \cos 30^\circ \\ -80 \sin 30^\circ \\ 0 \end{bmatrix} \text{ N}$

$\vec{W} = \begin{bmatrix} 0 \\ -196.2 \\ 0 \end{bmatrix} \text{ N}$

$\vec{F}_s = \begin{bmatrix} -F_s \\ 0 \\ 0 \end{bmatrix}$

$\vec{N} = \begin{bmatrix} 0 \\ N \\ 0 \end{bmatrix}$

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SUPPOSE IT'S IN EQ. $\Sigma \vec{F} = 0$

$$\begin{bmatrix} 80 \cos 30^\circ \\ -80 \sin 30^\circ \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -196.2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ N \\ 0 \end{bmatrix} + \begin{bmatrix} -F_s \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$80 \cos 30^\circ - F_s = 0 \Rightarrow F_s = 80 \cos 30^\circ \text{ N}$
 $-80 \sin 30^\circ - 196.2 + N = 0 \Rightarrow N = 236.2 \text{ N}$

$F_{s\max} = \mu_s \cdot N$
 $= (0.3)(236.2)$
 $= 70.86 \text{ N}$
 $F_s \leq F_{s\max}$

$\mu_s = 0.3$
 $F_{s\max} = (0.3)(236.2)$
 $= 70.86$
 $F_s > F_{s\max}$

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$\Sigma \vec{M}_C = 0$

$$\begin{bmatrix} -0.4 - x \\ 0.2 \\ 0 \end{bmatrix} \times \begin{bmatrix} 80 \cos 30^\circ \\ -80 \sin 30^\circ \\ 0 \end{bmatrix} + \begin{bmatrix} -x \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ -196.2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$
$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ (-x)(-196.2) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$-(0.2)(80 \cos 30^\circ) + (0.4 + x)(80 \sin 30^\circ) + (x)(196.2) = 0$

$x = \frac{-2.144}{80 \sin 30^\circ + 196.2} = \frac{-2.144}{236.2} = -0.00908 \text{ m}$

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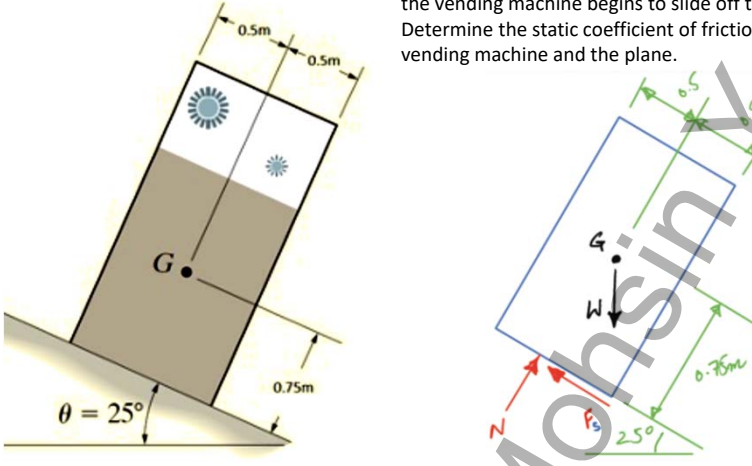
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Q. When the inclined plane is raised to an angle of $\theta = 25^\circ$, the vending machine begins to slide off the plane. Determine the static coefficient of friction between the vending machine and the plane.



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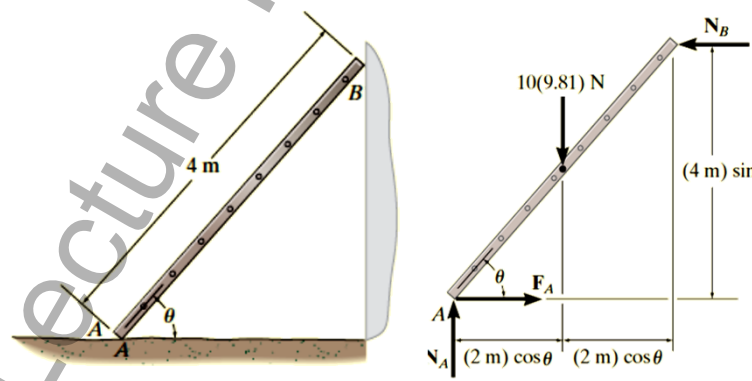
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The uniform 10-kg ladder in Fig. 8-9a rests against the smooth wall at B, and the end A rests on the rough horizontal plane for which the coefficient of static friction is $\mu_s = 0.3$. Determine the angle of inclination θ of the ladder and the normal reaction at B if the ladder is on the verge of slipping.



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Blocks *A* and *B* have a mass of 3 kg and 9 kg, respectively, and are connected to the weightless links shown in Fig. 8–11*a*. Determine the largest vertical force **P** that can be applied at the pin *C* without causing any movement. The coefficient of static friction between the blocks and the contacting surfaces is $\mu_s = 0.3$.

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FUNDAMENTAL PROBLEMS

FR-1. If $P = 200\text{ N}$, determine the friction developed between the 50-kg crate and the ground. The coefficient of static friction between the crate and the ground is $\mu_s = 0.3$.

FR-2. Determine the minimum force P to prevent the 30-kg rod *AB* from sliding. The contact surface at *B* is smooth, whereas the coefficient of static friction between the rod and the wall at *A* is $\mu_s = 0.2$.

FR-3. Determine the maximum force P that can be applied without causing the two 30-kg crates to move. The coefficient of static friction between each crate and the ground is $\mu_s = 0.25$.

FR-4. If the coefficient of static friction at contact points *A* and *B* is $\mu_s = 0.3$, determine the maximum force P that can be applied without causing the 100-kg spool to move.

FR-5. Determine the minimum force P that can be applied without causing movement of the 250-lb crate which has a center of gravity at *G*. The coefficient of static friction at the floor is $\mu_s = 0.4$.

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